

8/19/2026

Last Time:

Thm (Fundamental Thm of Arithmetic) Every $n > 1$ can be written as a product primes, uniquely up to the order of the prime factors.

Thm (Division Algorithm). Let $a, b \in \mathbb{Z}$, $b \neq 0$. Then there are unique $q, r \in \mathbb{Z}$ with $0 \leq r < |b|$, s.t.

$$a = q \cdot b + r.$$

$\uparrow \quad \uparrow$
quotient remainder

Pf. Exercise.

E.g. 1. $a = 100$, $b = 13$: $100 = 13 \cdot \underline{7} + \underline{9}$
 q r

$$2. a=100, b=-13 : 100 = -13 \cdot \underline{-7} + \underline{9}$$

$$3. a=-100, b=13 : -100 = 13 \cdot \underline{-8} + \underline{4}$$

Recall:

Lem 1.1. Every $n > 1$ has a prime divisor.

Pf of Existence part of FTA: Let $n > 1$. By Lem 1.1,

$$n = p_1 n_1, \quad p_1 \text{ is prime, } n > n_1.$$

$$n_1 = 1 : n = p_1 \cdot 1 = p_1 \quad \checkmark$$

$n_1 > 1$: By Lem 1.1 again,

$$n_1 = p_2 n_2, \quad p_2 \text{ is prime, } n_1 > n_2.$$

$$n_2 = 1 : n_1 = p_2 \cdot 1 = p_2, \quad n = p_1 n_1 = p_1 p_2. \quad \checkmark$$

$n_2 > 1$: Continue until we get

$$n = p_1 p_2 \cdots p_k \underline{n_k}, \quad n_{k-1} > n_k = 1.$$

We are done. □

Defn (Ideals) Let R be a comm ring. An ideal of R is

a set I s.t.:

(1) $0 \in I$

(2) $x, y \in I \Rightarrow x + y \in I$

(3) $r \in R, x \in I \Rightarrow rx \in I$.

E.g. (Ideals of $\mathbb{Z}$): $I = \{\text{even integers}\}$

$$I_m = \{n : m \mid n\}, \quad m \in \mathbb{Z} \text{ nonzero.}$$

Defn (Principal Ideal) An ideal I of R is called

principal if $I = \{r\alpha : r \in R\}$ for some $\alpha \in R$.

Defn (Principal Ideal Domain) A domain D is called a

PID if every ideal of D is principal.

Prop 2.1. $\mathbb{Z}$ is a PID.

Pf. Let I be an ideal. Need to show $I = \{n : m \mid n\}$ for

some m .

$$I = \{0\} : \checkmark$$

$I \neq \{0\}$: I contains some $m > 0$. We may assume m is the minimal positive element.

$$\text{Claim: } I = \{n : m|n\}.$$

$$\{n : m|n\} \subseteq I : \checkmark$$

$I \subseteq \{n : m|n\}$: Take any $n \in I$. By DA, can write

$$n = q \cdot m + r, \quad 0 \leq r < m.$$

$$r = n - \overbrace{qm}^{\in I} \in I \Rightarrow r = 0 \Rightarrow m|n. \quad \square$$

$\begin{array}{ccc} & \uparrow & \\ & I & \\ \uparrow & & \uparrow \\ I & & I \end{array}$

Notation : $\langle m \rangle = \{n : m|n\}$.

Defn (GCD) Let $a, b \in \mathbb{Z}$, not both 0. Then the gcd of

a, b vs a positive integer d s.t.

$$(1) \quad d|a, d|b$$

$$(2) \quad c|a, c|b \Rightarrow c|d.$$

We write $\gcd(a, b) = d$ or simply $(a, b) = d$.

Lem (Bézout's lemma) Let $a, b \in \mathbb{Z}$, not both 0. Write

$d = (a, b)$ and consider $I_{a,b} = \{ax + by : x, y \in \mathbb{Z}\}$. Then

$I_{a,b}$ is an ideal and $I_{a,b} = \langle d \rangle$. Consequently, there

are $x, y \in \mathbb{Z}$ s.t. $ax + by = d$.

Pf. $I_{a,b}$ is an ideal (Ex).

$$I_{a,b} \subseteq \langle d \rangle : d|a, d|b \Rightarrow d|ax + by \Rightarrow ax + by \in \langle d \rangle.$$

$\langle d \rangle \subseteq I_{a,b}$: By Prop 2.1, $I_{a,b} = \langle m \rangle$ for some $m \in \mathbb{Z}$.

Need to show $m|d$. Since $a, b \in I_{a,b}$, $m|a$, $m|b$.

So $m|d$. □

Lem (Euclid's lemma) Let $a, b \in \mathbb{Z}$ and let p be prime.

Then $p|ab \Rightarrow p|a$ or $p|b$.

pf. $p|a$ ✓

pfa: $(a, p) = 1$ (Ex),

Bézout's lemma $\Rightarrow ax + py = 1$, for some $x, y \in \mathbb{Z}$.

$$\Rightarrow abx + pby = b.$$

$p|ab$, $p|p \Rightarrow p|ab \cdot x + p \cdot by \Rightarrow p|b$. □